

SPARSE OBSERVATION ALGEBRAS, CONFLUENT DIRECTIONS, AND FINITE-STATE MEMORY

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ABSTRACT. For a positive finite detector spanning the monomials with exponents in A , we identify the generic dimension of the attainable information for n past and m future observations as $\min\{n(|A| - 1), |mA| - 1\}$. The formula follows from an attainable binomial product tangent and strict mixed-moment positivity for every full-support prior. We then determine what remains of this law when additive exponents coalesce. Divided differences yield a nonsingular pairing with logarithmic jets, while the actual query metric assigns successive powers of the collision parameter to these directions. In the full-future regime this gives a uniform, scale-sensitive checkpoint memory law determined by the cluster multiplicities. For the five-trial family $A_\theta = \{0, 1, 2 + \theta\}$ we prove the stronger streaming law

$$\mathcal{R}_{M,5}^\theta \asymp \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\},$$

uniformly in both the number M of persistent states and $0 \leq \theta \leq 1/2$. Its upper realization quantizes after every report, without division by θ ; its lower bound retains the probability of the acquired history. In one common binary decision task, erasing the weak direction loses order θ^2 , and a streaming controller recovers a fixed fraction of that value once M exceeds a constant times θ^{-4} . The general exact causal state, fixed-calibration control bounds, and a counted mechanical collision-bit comparison are also established.

1. INTRODUCTION

Exact distinguishability and distinguishability at a prescribed accuracy need not have the same dimension. An additive relation between observation exponents can disappear under an arbitrarily small perturbation, creating a new exact direction whose statistical amplitude tends to zero. Our object is to compute both the exact information law and its resolved finite-state counterpart through such a collision.

The likelihood accumulated by an experiment and the information needed for its remaining decisions are different objects. A posterior family can have many directions that no remaining experiment can test. Conversely, a sparse one-step detector can generate a much larger collection of future tests by multiplication. The appropriate question is therefore not only the rank of the detector, but the rank of the pairing between an *attainable product tangent* and the *future observation algebra*.

Here this pairing can be computed for a whole class of nonsaturated experiments. Let

$$A = \{0 = a_0 < a_1 < \cdots < a_{r-1} = D\}, \quad r \geq 2, \quad W_A = \text{span}\{t^a : a \in A\},$$

and let mA denote the set of sums of exactly m members of A , with $0A = \{0\}$. Thus zero-padding is allowed and $mA \subset (m+1)A$. Write $h_A(m) = |mA|$. The parameter lies in a fixed compact interval $I = [l, u]$ with $0 \leq l < u < \infty$, and its prior μ has full support. The detector

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and comparator are specified in Section 2. No function of the unknown parameter is supplied to the controller.

Theorem 1.1 (Sparse finite-horizon information law). *Suppose the positive finite detector spans W_A and every lookup entry ranges over $[\eta, 1 - \eta]$, where $0 < \eta < 1/2$. For $n, m \geq 0$ put*

$$(1) \quad d_A(n, m) = \min\{n(r - 1), h_A(m) - 1\}.$$

Two attainable posteriors after n reports have the same law under every common future-only continuation of length m if and only if their moments at the exponents in mA agree. The prediction map on all-failure command histories has generic rank $d_A(n, m)$, and it has a smooth attainable patch of this dimension with a continuous section to histories. The smallest Euclidean dimension of a sufficient encoding continuous on each known-report history stratum is exactly $d_A(n, m)$.

For a total budget N , these dimensions can be attained simultaneously by a causal exact-real state. Its maximal size is

$$(2) \quad D_A(N) = \max_{0 \leq n \leq N} d_A(n, N - n).$$

Only prior moments with exponents in NA are needed for its updates.

Generic rank here means rank outside a proper real algebraic exceptional set in the finite command coordinates. It is not a claim of uniform conditioning. The global upper encoding is an encoding of histories; when it stores normalized factors it need not identify different factorizations of the same posterior. We do not assert that every singular quotient has a global coordinate chart of its local dimension.

For $A = \{0, 1, \dots, q\}$, (1) becomes $q \min(n, m)$. For $A = \{0, 2, 5\}$ it instead gives $d_A(3, 2) = 5$. More generally, for $A = \{0, a, b\} \subset \mathbb{N}$ and $B = b / \gcd(a, b)$,

$$(3) \quad h_A(m) = \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2}.$$

For incommensurable $a, b > 0$, the same three-cell detector space has $h_A(m) = \binom{m+2}{2}$ at every m . Thus neither one-step rank nor maximum polynomial degree describes all the possible memory profiles.

The central step is a prior-independent transversality calculation, not an assumption that the relevant projection has maximal rank. At factors $1 + c_i t^D$, with the c_i distinct and small, the unnormalized tangent is the monomial span with exponent set

$$B_{A,n} = \{jD : 0 \leq j \leq n\} \cup \bigcup_{a \in A \setminus \{0, D\}} \{a + jD : 0 \leq j < n\}.$$

It has $n(r - 1) + 1$ members. Every square mixed-moment minor between this set and mA is strictly positive. Normalization removes exactly one rank. Section 3 supplies the full argument, including attainment inside one fixed comparator cube and the topological lower bound.

1.1. The uniform law at an additive collision. The exact formula is the starting point, rather than the end, of the finite-resolution analysis. Let the exponents depend affinely on a known calibration parameter, $A_\theta = \{a_i + \theta b_i\}$. Group the formal m -fold sum exponents by their limiting value at zero. A cluster of multiplicity s contributes orders $0, 1, \dots, s - 1$. After deleting the constant coordinate, list all orders as $\nu_1 \leq \dots \leq \nu_d$, with d the nearby future dimension. Theorem 5.1 proves, whenever $d \leq n(r - 1)$,

$$(4) \quad R_{M;n,m}^\theta \asymp \max_{1 \leq \ell \leq d} \theta^{2(\nu_1 + \dots + \nu_\ell)/\ell} M^{-2/\ell}$$

for both unconditional and minimax checkpoint prediction, with constants uniform through the collision. The calibration, prior and finite horizon are fixed. Distinct positive collision parameters use the same command and query labels. The observation of a logarithmic jet is not postulated: Newton interpolation identifies its actual coefficient in the physical query vector. Strict confluent positivity proves that the desingularized attainable patch survives at zero, including for priors without a density.

The family $A_\theta = \{0, 1, 2 + \theta\}$ supplies a complete sequential realization of this law. In the positive three-cell model (29), with total horizon five, its exact peak is four at zero and five for every $0 < \theta \leq 1/2$. Nevertheless the new direction has width θ , not unit width. Theorem 7.1 proves

$$(5) \quad \overline{\mathcal{R}}_{M,5}^{\theta, \text{stream}} \asymp \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}$$

uniformly for every integer $M \geq 1$ and $\theta \in [0, 1/2]$. The same upper filter works on every admitted history and at every checkpoint. The transition from a four-dimensional to a five-dimensional rate occurs at order $M = \theta^{-4}$, or distortion θ^2 . This determines, rather than leaves inside an unspecified constant, the degeneration of the high-resolution law.

A singular coordinate system would not suffice to prove (5): a transition with norm $1/\theta$ could amplify an earlier quantization error and destroy uniformity. The proof uses normalized factors before the peak, four physical moments and one unscaled moment difference at the peak, and two moments afterward. Its shrinking update (37) contains no division by θ . Reachable representatives therefore give a genuine index-only filter with uniformly accumulated errors.

Finally, after the same exploration and query, offer a ticket on the same future event at an independent uniform price. A binary accept-or-reject action then has optimal value gap exactly half the squared-prediction regret. More substantially, Theorem 8.2 compares one common payoff before and after the deterministic history statistic T_θ discards the weak coordinate. The full loss is comparable to θ^2 uniformly, and a streaming controller with $M \geq K\theta^{-4}$ beats even the uncompressed erased statistic by at least a fixed positive multiple of θ^2 . This is not an assertion that every four-dimensional encoding is inadequate, nor an optimal-exploration theorem. It gives the resolved sparse direction a quantitative value in one precisely specified sequential decision.

1.2. Fixed-calibration consequences and mechanical realization. An exact-real causal formula alone does not give a finite-state controller. Section 6 constructs transition maps that store only one of M symbols after *every* report. For fixed $N \geq 2$, its prediction regret is at most $C_N M^{-2/D_A(N)}$ at any checkpoint. At a checkpoint maximizing (2), one fixed randomized exploration experiment has regret at least $c_N M^{-2/D_A(N)}$ for every such finite-state filter, including the declared independent randomization. These bounds match the exponent, not the leading distortion constant. The query is announced only after the checkpoint state has been formed, and old commands or command-generation seeds cannot be retained outside that state.

The lower bound concerns a prescribed exploration-and-prediction task; it is not a universal lower bound on all optimally chosen control tasks. For bounded additive observable rewards we separately prove an exact-model $C'_N M^{-1/D_A(N)}$ optimal-control loss bound. We keep that weaker power distinct from the squared-prediction power. For an all-failure probe whose factors are bounded by $v < 1$, a one-state zero forecast has regret at most v^{2m} . That elementary finite-resolution bound remains visible alongside the new streaming rate.

Section 9 studies the collision bit using one common prediction target and one common total-risk baseline. The optimum for *every* memory budget is given by a finite partition formula. Four states realize the full strict gain. For a sufficiently small positive amplitude, three states

have exactly the same optimal risk as the erased experiment, even though the exact detector rank is already larger. This identifies an operational resolution effect rather than equating rank with useful accuracy.

1.3. Relation to earlier work. Action-conditional predictions as representations of state are classical [6]. Approximate recursively updated information states and their dynamic-programming loss bounds are treated in [8]; finite-window policies under filter stability are studied in [5]. Squared-error quantization supplies the covering exponent [4]. Total positivity and generalized Vandermonde matrices are classical [7]. The identification of iterated sumsets with Hilbert functions is also established [2, 3]; it is not a novelty claim here.

The result proved here concerns their specific junction: the attainable normalized-product tangent of a sparse calibrated experiment pairs at maximal rank with its future test space for every full-support prior. This yields the explicit law (1), its causal realization, and the finite-state consequences with the same attainable geometry. None of the cited general representation, sumset, positivity or quantization principles alone is used as a substitute for this calculation. The complete v6 predecessor is retained [13]. The full-space predecessor and its mechanical derivations are retained as a complete source companion [9]. The sparse three-cell example that motivated the present generalization was supplied in the source-pinned v5 referee report [10]; its authorship is not reassigned here.

The additional quantitative input is the uniform attainable pairing with confluent exponent jets and its transfer to actual sequential state updates. The divided-difference identities and rectangular covering calculation are classical; they do not by themselves establish attainable surjectivity at the singular calibration, or prevent error amplification in a causal implementation. The near-collision example and its preliminary base-model bound originate in [14]. Equations (4)–(5) and the same-task weak-direction value are the assertions whose proofs occupy Sections 5, 7 and 8.

2. CALIBRATED EXPERIMENTS AND THEIR OPERATIONAL QUOTIENT

All spaces carry their Borel sigma-algebras. Deterministic continuation, transition and decision rules are Borel measurable; randomized rules are Borel probability kernels. Every external random seed is independent of the hidden parameter and the acquired history unless explicitly charged as part of that history.

Let $k_1, \dots, k_K \in W_A$ be strictly positive on I , satisfy $\sum_j k_j = 1$, and span W_A . On each trial a fresh raw symbol has these probabilities conditional on the same fixed parameter t . The raw trials are conditionally independent. A command $g \in \mathcal{G} = [\eta, 1 - \eta]^K$ accepts raw symbol j with probability g_j , using an independent prepared uniform variable. A rejected symbol is reported as $\dagger$. Thus the report alphabet is $\mathcal{X} = \{1, \dots, K, \dagger\}$ and its likelihood factors are

$$(6) \quad f_{g,j} = g_j k_j, \quad f_{g,\dagger} = F_g = \sum_j (1 - g_j) k_j.$$

A single command is applied at every possible parameter value. Every trial is counted, including every rejected report. There is a constant $\kappa > 0$ such that $\kappa \leq f_{g,x}(t) \leq 1$ for all g, x, t . Indeed one may take $\kappa = \eta \min_{j,t} k_j(t)$.

Lemma 2.1 (Attainable interior and realizability). *The failure family is precisely the image of the rejection cube under $u \mapsto \sum_j u_j k_j$. It contains a neighborhood of the constant $1/2$ in W_A . Conversely, every finite exponent set A as above has a positive normalized r -cell realization with this property.*

Proof. The displayed map is a surjective linear map onto W_A . Choose a linear right inverse $L : W_A \rightarrow \mathbb{R}^K$. For every v satisfying $\|Lv\|_\infty < 1/2 - \eta$, rejection vector $u = \mathbf{1}/2 + Lv$ realizes $1/2 + v$ inside the cube. This proves both the interior statement and that the same physical command implements it. For existence, choose $\delta > 0$ with $\delta \sum_{a \in A \setminus \{0\}} u^a < 1/r$ and set

$$k_a(t) = 1/r + \delta t^a \quad (a \neq 0), \quad k_0(t) = 1/r - \delta \sum_{a \neq 0} t^a.$$

They are positive and sum to one. Their span contains one and then each $t^a = (k_a - 1/r)/\delta$, so has dimension r . These are calibrated categorical experiments. No mechanical origin is asserted for this general realization; the separate Lorentz realization is derived in Section 9. $\square$

A history $h = (g_1, x_1, \dots, g_n, x_n)$ has posterior

$$(7) \quad P_h(t) \mu(dt), \quad P_h = \frac{\prod_{i=1}^n f_{g_i, x_i}}{\mu(\prod_{i=1}^n f_{g_i, x_i})}.$$

All commands are known inputs when received. They cease to be available once discarded by a memory-limited filter. A continuous history encoding means continuity in $(g_1, \dots, g_n)$ separately for each fixed report word. It need not be injective on histories and need not identify all histories with the same posterior.

Definition 2.2 (Common future-only continuation). At a checkpoint a continuation has the same fixed task data in the two experiments. Its commands are functions only of the reports and commands *since that checkpoint*, and any fresh independent random seed. Its first command is the same in both experiments. It may not read the original prefix, a prefix-dependent seed, or any other uncharged historical tape. Two posteriors are m -equivalent when all such length- m continuations have the same future report law.

A separate payoff automaton referring to past observations is task memory, not something removed by posterior equivalence. Losses that inspect t directly are also not included without adding their test functions. This quantifier matters: two different constant failure commands leave the same prior, but a rule that reads and repeats the original command does not meet Definition 2.2.

Proposition 2.3 (Exact future test space). *The linear span of length- m future likelihood tests is*

$$(8) \quad W_A^m = \text{span}\{t^s : s \in mA\}.$$

Two posteriors are m -equivalent if and only if their moments at the exponents in $mA \setminus \{0\}$ agree. A finite list of actually executable, nonadaptive all-failure probes spans the same space.

Proof. Choose finitely many attainable failures spanning W_A , using Lemma 2.1. All their m -fold products are executable all-failure event probabilities. Multilinearity makes their span the span of all products of m members of W_A , which is exactly (8). Independence is conditional on t ; one does not replace the shared parameter by fresh parameters between trials.

For any fixed future word a deterministic continuation chooses a known sequence of commands along that word. Its likelihood is a product of m members of W_A , even when the chosen sequence differs on other branches. Equality of the stated moments therefore implies equality of all word probabilities. There are finitely many report words. Integrating an independent continuation seed proves the randomized case. Conversely, equivalence includes all the spanning all-failure probes; their predictions determine all the moments in (8). The constant moment is one. Only the selected probe is executed and all its m trials are consumed, regardless of earlier nonfailure reports. $\square$

A fixed positive nonconstant denominator or an arbitrary nonmonomial factor space is not silently substituted in (8). Such models have their own product and test pairings. The theorem below addresses every sparse monomial space W_A , not every finite-dimensional space of continuous functions.

3. ATTAINABLE TRANSVERSALITY AND THE EXACT MEMORY PROFILE

We first prove the positivity statement that prevents a prior-dependent loss of rank. No density assumption on the prior is needed.

Lemma 3.1 (Strict mixed-moment positivity). *Let $\alpha_1 < \dots < \alpha_s$ and $\beta_1 < \dots < \beta_s$ be nonnegative real numbers. For every full-support probability on I ,*

$$(9) \quad \det[\mu(t^{\alpha_i + \beta_j})]_{i,j=1}^s > 0.$$

Consequently a rectangular moment matrix indexed by two increasing finite sets of distinct nonnegative exponents has rank the smaller cardinality.

Proof. For increasing positive points $t_1 < \dots < t_s$, the generalized Vandermonde determinant $\det[t_j^{\alpha_i}]$ is positive. Here is a self-contained sign argument. Put $t_j = e^{x_j}$. A nonzero combination of s exponentials with distinct ordered exponents has at most $s - 1$ distinct zeros: factor out the first exponential, differentiate, and use Rolle's theorem inductively. Thus the determinant cannot vanish on the connected region $x_1 < \dots < x_s$. When the x_j coalesce to x_0 from ordered nearby points, its first nonzero Taylor term has the sign of the Wronskian

$$e^{x_0 \sum_i \alpha_i} \prod_{i < j} (\alpha_j - \alpha_i)$$

times the ordinary positive Vandermonde in the x_j , divided by positive factorials. Its sign is therefore positive everywhere in that region. The same holds for the β determinant.

Expanding determinants and applying Fubini gives the finite determinant integration identity

$$\det[\mu(t^{\alpha_i + \beta_j})] = \frac{1}{s!} \int_{I^s} \det[t_j^{\alpha_i}] \det[t_j^{\beta_i}] \prod_{j=1}^s \mu(dt_j).$$

The integrand is nonnegative, by permutation symmetry and continuity at zero or repeated points. Choose s disjoint ordered subintervals in $(l, u) \cap (0, \infty)$. Each has positive prior mass by full support; on their product the integrand is strictly positive. The integral is strictly positive. Applying this to any square minor proves the last assertion. This is the classical total-positivity mechanism [7], with its needed form proved explicitly here. $\square$

Lemma 3.2 (A monomial tangent inside the attainable cube). *For every $n \geq 1$, choose distinct, sufficiently small positive numbers $c_1, \dots, c_n$. The failure factors $F_i = (1 + c_i t^D)/2$ are simultaneously attainable at interior commands. At this tuple the differential image of the unnormalized product map $W_A^n \rightarrow \text{span}\{t^s : s \in nA\}$ is*

$$(10) \quad T_{A,n} = \text{span}\{t^b : b \in B_{A,n}\},$$

where

$$(11) \quad B_{A,n} = \{jD : 0 \leq j \leq n\} \cup \bigcup_{a \in A \setminus \{0, D\}} \{a + jD : 0 \leq j < n\}.$$

In particular $\dim T_{A,n} = n(r - 1) + 1$, and the product itself belongs to this tangent space.

Proof. Attainability follows by applying the fixed right inverse in Lemma 2.1 to $c_i t^D/2$ and taking all c_i small enough. The factors and their implementing commands lie in an open subset of the allowed cube. Constants 2^{-n} do not change a differential image, so omit them in the remaining calculation. Put $z = t^D$, $P(z) = \prod_i (1 + c_i z)$ and $P_i(z) = P(z)/(1 + c_i z)$. These n polynomials form a basis of the polynomials in z of degree at most $n - 1$. Indeed evaluation at $z = -1/c_j$ makes all P_i except P_j zero, and $P_j(-1/c_j) \neq 0$.

The product differential is $\sum_i \dot{F}_i \prod_{j \neq i} F_j$. Variations in the constant direction give all polynomials in z through degree $n - 1$; variations in t^D give their multiples by z . Together they give all z^j , $0 \leq j \leq n$. Variations in an interior exponent a give precisely $t^a z^j$, $0 \leq j < n$. The listed exponent sets are disjoint: two numbers $a + jD$ and $a' + j'D$ with $0 < a, a' < D$ agree only if $a = a'$ and $j = j'$, and none is a multiple of D . Their total cardinality is $(n + 1) + n(r - 2) = n(r - 1) + 1$. Every summand of the differential has this form, proving equality rather than only inclusion in (10). The product is obtained by varying one factor along itself, so is in the tangent. $\square$

Theorem 3.3 (Generic normalized pairing rank). *For fixed $n, m \geq 1$, the map from n all-failure commands to the moment vector $(\mu(t^s P_h))_{s \in mA \setminus \{0\}}$ has generic rank $d_A(n, m)$ in (1). This holds for every full-support prior, including priors without a density. At an interior attainable tuple it has a smooth $d_A(n, m)$ -dimensional local image and a continuous local section to histories. A projection onto $d_A(n, m)$ of the moment or spanning-probe coordinates maps such a section onto an open Euclidean ball.*

Proof. Let $P = \prod_i F_i$, $Z = \mu P > 0$, and include the constant test in the linear map

$$L : T_{A,n} \longrightarrow \mathbb{R}^{h_A(m)}, \quad LQ = (\mu(t^s Q))_{s \in mA}.$$

At the tuple from Lemma 3.2, Lemma 3.1 gives

$$\text{rank } L = \min\{n(r - 1) + 1, h_A(m)\}.$$

Let e_0 extract the coordinate of exponent zero, and put $p = L(P)/Z$. The derivative of the normalized prediction map on this tangent is

$$(12) \quad Q \longmapsto Z^{-1}(LQ - p e_0(LQ)).$$

Now $e_0 p = 1$ and $p \in L(T_{A,n})$. The linear map $v \mapsto v - p e_0 v$, restricted to $L(T_{A,n})$, has kernel exactly $\text{span}\{p\}$ and image in the zero-constant-coordinate hyperplane. It loses *exactly* one rank. Deleting that identically zero coordinate therefore gives $\text{rank } \min\{n(r - 1), h_A(m) - 1\}$.

At every tuple the normalized product map is invariant under independent rescaling of its n nonzero factors, so its rank is at most $n(r - 1)$. The number of nonconstant tests gives the other upper bound. Thus the attained rank is globally maximal. The command-to-factor linear map is surjective at every interior command; the same maximal rank is attained in command coordinates. Its Jacobian entries are rational functions of the commands with strictly positive polynomial denominator Z^2 . A maximal minor nonzero at the displayed tuple has a nonzero polynomial numerator. Its zero set is a proper real algebraic set with empty interior and zero Lebesgue measure. This proves genericity throughout the command cube, not only along the binomial family.

Choose $d = d_A(n, m)$ independent output coordinates and d command coordinates at that tuple, and freeze the complementary command coordinates. The inverse function theorem gives a local inverse for the selected output projection; shrink to a ball with compact closure. The full image of this section is a smooth local graph. A spanning list of probes is an injective linear image of the moment coordinates, so the same conclusion holds after choosing independent

probe coordinates. All section histories are feasible all-failure histories. Their probability conditional on each parameter is at least η^n ; no failure evidence has been dropped. $\square$

Notice that when $d < h_A(m) - 1$ the attainable image can be curved. The proof does not replace it by its affine hull. The lower-bound ball is in a selected output projection, and its lift is a local graph, not necessarily a flat ball in the ambient prediction space.

Theorem 3.4 (Minimal continuous encoding and causal attainment). *The minimal dimension stated in Theorem 1.1 is $d_A(n, m)$. For a fixed N , there are compact exact state sets $\mathcal{S}_n \subset \mathbb{R}^{d_A(n, N-n)}$ and continuous-in-command causal updates attaining this dimension at every n . Their state represents the posterior-dependent future law, not an uncharged past-dependent task.*

Proof. Compose a sufficient continuous encoding with the section in Theorem 3.3. Different points in the projected ball have different predictions, so this composition is injective. A continuous injection of an open d -ball into $\mathbb{R}^k$, $k < d$, is impossible: embedding $\mathbb{R}^k$ as a coordinate plane of $\mathbb{R}^d$ would contradict invariance of domain. The decoder need not be continuous for this argument.

For the global upper bound, normalize each observed factor by its own prior integral, $\tilde{f}_i = f_i / \mu f_i$. Each normalized factor is in the affine hyperplane $\mu f = 1$ of W_A , with $r - 1$ coordinates. The chronologically ordered n factors determine the posterior product and use $n(r - 1)$ real coordinates. This representation may distinguish permutations; sufficiency does not forbid that. Alternatively, retain the $h_A(m) - 1$ nonconstant posterior moments in mA . Proposition 2.3 proves this alternative sufficient on every history. Use whichever representation is smaller. Both are continuous on each fixed-report stratum. Finite unions of compact continuous images of the command cube give compact state sets. The cases $n = 0$ or $m = 0$ have a single class and zero coordinates.

For causal implementation retain normalized factors while $n(r - 1) \leq h_A(N - n) - 1$. The left side increases and the right side nonincreases, so there is at most one changeover. When the inequality first fails, multiply the retained factors and compute the required moments. Afterward, if $m = N - n$ and the new report factor is $f(t) = \sum_{a \in A} f_a t^a$, update by

$$(13) \quad M'_s = \frac{\sum_{a \in A} f_a M_{s+a}}{\sum_{a \in A} f_a M_a}, \quad s \in (m - 1)A \setminus \{0\}, \quad M_0 = 1.$$

All indices $s + a$ belong to mA . The denominator is the report probability and is at least κ . At the changeover the integrands are products of n factor monomials and a remaining-test monomial, so their exponents are in $nA + (N - n)A = NA$. Normalizing each factor requires only moments in $A \subset NA$. Thus a finite read-only table $(\mu(t^s))_{s \in NA}$ suffices for all updates. Formula (13) then gives the exact shrinking state to time N . $\square$

For arbitrary nonmonomial spaces W , the starting identity remains (12), with the true product tangent and W^m . What is special here is the explicit attainable tangent (10) and the strictly positive mixed pairing. No dimension formula depending only on $\dim W$ is inferred outside this class.

4. THE OBSERVATION ALGEBRA AND ASYMMETRIC PROFILES

For integral exponents define the standard graded algebra

$$\mathcal{R}_A = \mathbb{R}[zt^a : a \in A] \subset \mathbb{R}[z, t], \quad \deg(zt^a) = 1.$$

Its degree- m component has basis $\{z^m t^s : s \in mA\}$, so its Hilbert function is exactly $h_A(m)$. The general sumset–Hilbert-function connection is classical [2, 3]. Its role here is operational:

$h_A(m) - 1$ is the number of independent future observable tests, and Theorem 3.3 proves when the attainable past can excite all of them. The algebra alone would not prove that transversality.

Proposition 4.1 (Three-cell classification). *Let $A = \{0, a, b\}$ with integers $0 < a < b$, put $g = \gcd(a, b)$, $a' = a/g$ and $B = b/g$. Then*

$$(14) \quad h_A(m) = \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2}.$$

The homogeneous relation is generated by $x_1^B - x_0^{B-a'} x_2^{a'}$ under the map $(x_0, x_1, x_2) \mapsto (z, zt^a, zt^b)$. In particular,

$$d_A(n, m) = \min \left\{ 2n, \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2} - 1 \right\}.$$

For nonnegative real exponents $0 < a < b$ with a/b irrational, $h_A(m) = \binom{m+2}{2}$ for every m .

Proof. A monomial $x_0^k x_1^i x_2^j$ of degree m maps to $z^m t^{ai+bj}$. Two exponent triples of the same degree have the same image exactly when their difference is an integer multiple of $-(B - a'), B, -a'$: this follows by dividing $a(i - i') + b(j - j') = 0$ by g and using coprimality of a', B . After cancelling a common monomial, each corresponding binomial is a multiple of $x_1^B - x_0^{B-a'} x_2^{a'}$, by the difference-of-powers identity. The kernel is spanned by these binomials, because monomials with distinct images are linearly independent and each fiber's coefficient sum must vanish. Thus this one relation generates the kernel.

The degree- m polynomials in three variables have dimension $\binom{m+2}{2}$. Multiplication by the displayed nonzero degree- B relation is injective in the polynomial ring, so the degree- m part of the ideal has dimension zero for $m < B$ and $\binom{m-B+2}{2}$ otherwise. Subtraction proves (14). If a/b is irrational, $a(i - i') + b(j - j') = 0$ forces both differences to vanish; all $\binom{m+2}{2}$ possible pairs are distinct. The operational formula now follows from Theorem 3.3. $\square$

Corollary 4.2 (The sparse space that separates rank from future dimension). *For $A = \{0, 2, 5\}$, the first values of $h_A(m)$, starting at $m = 0$, are 1, 3, 6, 10, 15, 20, 25. At $n = 3, m = 2$ the exact information dimension is five, for every full-support prior. For a total budget of six the minimal causal dimensions, starting at $n = 0$, are*

$$0, 2, 4, 6, 5, 2, 0.$$

Thus the peak need not be the peak of a rank-based symmetric tent.

Proof. Apply (14) with $B = 5$ and then (1). In particular $2A = \{0, 2, 4, 5, 7, 10\}$, while the binomial witness at $n = 3$ has $B_{A,3} = \{0, 2, 5, 7, 10, 12, 15\}$. The mixed matrix has rank six and normalization leaves five. This proves the example uniformly over full-support priors without relying on a single large rational determinant. The uniform-prior three-cell example and its numerical witness first appeared in the source-pinned report [10]; the present corollary is an application of the general theorem, not a new attribution of that example. $\square$

There is also an exact finite computation for any integral A : start with $S_0 = \{0\}$, form $S_{m+1} = \{s + a : s \in S_m, a \in A\}$, and count distinct exponents. This handles all additive collisions, not just a nominal degree range. For real exponents the formula is equally exact as mathematics, but an effective implementation requires exact equality information about the supplied exponents. Neither fixed rank nor arbitrarily accurate floating-point exponents substitute for that information.

5. CONFLUENT OBSERVATION ALGEBRAS AND RESOLVED DIRECTIONS

The cardinality of a sumset is discontinuous when two of its elements coalesce. The corresponding experiment need not be discontinuous. We now identify the scales of the directions lost at such a collision. The parameter θ below indexes a *known calibration*; the unknown parameter of each experiment is still t .

5.1. Affine exponent families. Fix $0 = a_0 < a_1 < \cdots < a_{r-1}$ and real numbers b_i , with $b_0 = 0$. On a sufficiently small compact interval $0 \leq \theta \leq \theta_0 \leq 1$ put

$$A_\theta = \{a_i + \theta b_i : 0 \leq i < r\}.$$

The order of these exponents is assumed unchanged. Take $I = [0, 1]$ and one fixed full-support prior μ . A fixed coefficient matrix specifies positive detector cells

$$(15) \quad k_j^\theta(t) = \sum_{i=0}^{r-1} c_{ji} t^{a_i + \theta b_i}, \quad \sum_j k_j^\theta = 1.$$

It has column rank r , and the cells have a common positive lower bound. The comparator cube is fixed. These hypotheses include any sufficiently small affine perturbation of a strictly positive calibration of full rank. No continuity of an unknown prior density is assumed.

For a fixed future length m , let

$$\mathcal{B}_m = \left\{ \left(\sum_{\nu=1}^m a_{i_\nu}, \sum_{\nu=1}^m b_{i_\nu} \right) : 0 \leq i_\nu < r \right\}.$$

Repeated *pairs* are identified. Distinct pairs give distinct exponents $\lambda + \theta\gamma$ for $0 < \theta \leq \theta_0$; shrink θ_0 to exclude crossings if necessary. For each first coordinate λ , order its second coordinates as $\gamma_{\lambda,0} < \cdots < \gamma_{\lambda,s_\lambda-1}$. Set $K_m = \sum_\lambda s_\lambda$. The cluster at $\lambda = 0$ consists only of $(0,0)$.

Define the divided-difference functions

$$(16) \quad \psi_{\lambda,k}^\theta(t) = [\lambda + \theta\gamma_{\lambda,0}, \dots, \lambda + \theta\gamma_{\lambda,k}](x \mapsto t^x), \quad 0 \leq k < s_\lambda.$$

At $\theta = 0$ their continuous values are

$$(17) \quad \psi_{\lambda,k}^0(t) = t^\lambda (\log t)^k / k!.$$

For $\lambda > 0$ these functions have value zero at $t = 0$; the sole zero-exponent function is the constant one. The bracket in (16) is a divided difference in the exponent, not a new apparatus output. For a history h let $z_\theta(h)$ be its posterior expectations of these functions, omitting the constant coordinate.

List the orders k of the nonconstant functions in nondecreasing order:

$$0 = \nu_1 \leq \cdots \leq \nu_d, \quad d = K_m - 1.$$

There are $|mA_0| - 1$ zero orders. The associated resolution function is

$$(18) \quad \Phi_\theta(M) = \max_{1 \leq \ell \leq d} \theta^{2(\nu_1 + \cdots + \nu_\ell)/\ell} M^{-2/\ell}, \quad M \geq 1.$$

We set $0^0 = 1$ and $0^a = 0$ for $a > 0$ in this formula. In particular $\Phi_0(M) = M^{-2/(|mA_0|-1)}$.

Choose r attainable failure factors which form a basis in the formal monomials $t^{a_i + \theta b_i}$, with coefficients independent of θ . For example use $1/2$ and $1/2 + \delta t^{a_i + \theta b_i}$, $i > 0$, with δ sufficiently small. The fixed right inverse of the calibration matrix realizes these by fixed command labels. All ordered m -fold products form a common finite probe menu. Any other such fixed basis, and its full product menu, gives equivalent bounds below.

Theorem 5.1 (Uniform resolved checkpoint law). *Suppose*

$$(19) \quad K_m - 1 \leq n(r - 1).$$

Supply independent uniform commands for the first n trials, then reveal an independent uniform query from the preceding product menu. Let $\bar{R}_{M;n,m}^\theta$ be the optimal unconditional squared-prediction regret of an arbitrary M -message checkpoint encoder. Let $R_{M;n,m}^{\theta,\max}$ be its minimax counterpart over admitted histories. There are constants $0 < c \leq C < \infty$, independent of M and θ , such that

$$(20) \quad c\Phi_\theta(M) \leq \bar{R}_{M;n,m}^\theta \leq R_{M;n,m}^{\theta,\max} \leq C\Phi_\theta(M), \quad 0 \leq \theta \leq \theta_0.$$

The encoders may be Borel and randomized as in Definition 6.1. At the common interior binomial command tuple the normalized map $h \mapsto z_\theta(h)$ is a submersion of rank d , including at $\theta = 0$. Its local inverse bounds and an unconditional minorization can be chosen uniformly in θ .

The upper bound in this theorem is a checkpoint bound. It does not assume that access to an exact prefix is free during repeated compression. Every streaming filter is a checkpoint encoder, so the lower bound applies to such filters as well. Section 7 proves a matching genuinely streaming bound across an entire singular family.

5.2. Confluent positivity and the physical prediction metric. We give the details of the analytic and geometric steps in Theorem 5.1.

Lemma 5.2 (Strict confluent mixed pairing). *Let $\alpha_1 < \dots < \alpha_s$ be nonnegative, and let λ run through distinct nonnegative numbers with multiplicities q_λ summing to s . Include the columns $t^\lambda(\log t)^k/k!$, $0 \leq k < q_\lambda$, in this order within each increasing λ cluster. Assume $q_0 = 1$ if zero occurs. The matrix of their integrals against $t^{\alpha_i}\mu(dt)$ has strictly positive determinant. Consequently the corresponding rectangular pairing has full column rank whenever there are at least as many distinct row exponents as columns.*

Proof. Put $t = e^x$. The functions become $e^{\lambda x}x^k/k!$. A nonzero linear combination with total multiplicity s has at most $s - 1$ distinct zeros. Indeed divide by the first exponential and differentiate q_{λ_1} times. Rolle's theorem reduces a hypothetical s zeros to at least $s - q_{\lambda_1}$ zeros. The remaining terms are exponential polynomials with distinct nonzero exponents and the same remaining polynomial degrees: the operator $(D + \lambda - \lambda_1)^{q_{\lambda_1}}$ is invertible on polynomials of each fixed degree bound. Induction gives a contradiction. If no remaining term occurs, the original function is an exponential times a polynomial of degree less than s , and the assertion is immediate.

Thus the evaluation determinant at distinct ordered points never vanishes. Its sign is positive. One can determine the sign by letting ordered simple exponents approach the prescribed repeated exponents and dividing by their positive within-cluster Vandermonde factors. Equivalently, the Wronskian of the displayed exponential-polynomial basis has sign

$$\prod_{\lambda < \lambda'} (\lambda' - \lambda)^{q_\lambda q_{\lambda'}} > 0;$$

the exponential factor and the factorial normalizations are positive. Nonvanishing fixes this sign throughout the connected ordered-point domain.

Use the determinant integration identity from Lemma 3.1, with one determinant replaced by this confluent evaluation determinant. Both determinants have the same positive sign on ordered distinct positive points. The integrand is nonnegative elsewhere by symmetry and continuity. Separated ordered subintervals have positive prior mass, so the integral is strictly

positive. The logarithmic functions are bounded at zero under the stated hypotheses, which also justifies Fubini. Select any s rows for the rectangular assertion. $\square$

Lemma 5.3 (Desingularization and observable scales). *The functions in (16), and the needed derivatives in θ , are uniformly bounded on $[0, 1] \times [0, \theta_0]$. The posterior vectors $z_\theta(h)$ therefore lie in a fixed box. For the common product probe menu let $p_\theta(h)$ be its probability vector. There is a fixed matrix G of full column rank such that*

$$(21) \quad p_\theta(h) - p_\theta(\tilde{h}) = G \operatorname{diag}(\theta^{\nu_i})(z_\theta(h) - z_\theta(\tilde{h})).$$

In particular the squared norm on the left is bounded above and below by positive constants times $\sum_i \theta^{2\nu_i} |z_i(h) - z_i(\tilde{h})|^2$, with constants independent of θ .

Proof. The divided difference of order k is the integral of $f^{(k)}(\sum_j w_j x_j)$ over the standard k -simplex, whose volume is $1/k!$. This identity follows by iterating the fundamental theorem of calculus, starting with $(f(y) - f(x))/(y - x) = \int_0^1 f'(x + s(y - x)) ds$. It also gives the repeated-node value. Here $f(x) = t^x$, so every integrand is $t^x (\log t)^k$. At a positive cluster the exponent is bounded below by some $a > 0$ uniformly on the fixed parameter interval. For every fixed j , $\sup_{0 < t \leq 1} t^a |\log t|^j < \infty$. The same domination applies after finitely many θ derivatives. There are no higher divided differences at zero. Posterior expectations of bounded functions are bounded independently of the history.

Newton's interpolation identity, proved by successive subtraction of the interpolating polynomials, is exact at each node:

$$(22) \quad t^{\lambda + \theta \gamma_{\lambda,j}} = \sum_{k=0}^j \theta^k \left(\prod_{v < k} (\gamma_{\lambda,j} - \gamma_{\lambda,v}) \right) \psi_{\lambda,k}^\theta(t).$$

The matrix multiplying $\operatorname{diag}(\theta^k)$ is fixed, triangular within each cluster, and has nonzero diagonal. The standard identities used here are classical divided-difference calculus [12]; the integral proof makes the uniformity at coalescence explicit.

The expansion matrix from the formal pair monomials in $\mathcal{B}_m$ to the probe products has full column rank. To see this, invert the fixed one-step basis change and multiply it m times. Every formal monomial product is a linear combination of the probe products. Identifying equal pairs is a surjective linear quotient, so the spanning statement survives that identification. Both this expansion matrix and the triangular matrix in (22) are independent of θ . Their product has full column rank. Subtracting two normalized posterior vectors removes its constant column and gives (21). The remaining columns are still independent. Their extreme singular values give the two constants. The equality extends to $\theta = 0$ by continuity; vanishing weights then remove precisely the unresolved jets. $\square$

Lemma 5.4 (Uniform attainable patch with its evidence). *Under (19), there are $\rho, \beta > 0$, independent of θ , such that the subprobability distribution of z_θ on all-failure prefixes dominates β times uniform probability on a translate of $[-\rho, \rho]^d$.*

Proof. Use the binomial factors $(1 + c_i t^{D_\theta})/2$ of Lemma 3.2, with distinct small c_i and $D_\theta = a_{r-1} + \theta b_{r-1}$. Since the coefficient calibration is fixed, one common interior command tuple implements this family for every θ . Its unnormalized tangent has $n(r-1) + 1$ distinct monomials, also at zero. At positive θ , pairing with the K_m distinct future exponents has rank K_m by Lemma 3.1. The divided-difference basis change is invertible there. At zero, Lemma 5.2 gives the same rank directly, without passing a rank assertion through a singular limit. The product itself lies in the tangent. The normalization map $v \mapsto v - p_{e_0} v$ has kernel exactly $\operatorname{span}\{p\}$ on this full image, as in (12). Thus the normalized jet map has rank d at every θ , including zero.

Here are uniform local details. Let q be the total number of command coordinates and g_* the common interior tuple. The matrices $J_\theta = D_g z_\theta(g_*)$ are continuous in θ and surjective. Their least row singular value has a positive minimum σ on the compact parameter interval. Choose an orthonormal basis E_θ of the row space and a complementary orthonormal basis F_θ of its kernel, and use coordinates $g = g_* + E_\theta u + F_\theta w$. The derivative at zero of

$$(u, w) \mapsto (z_\theta(g_* + E_\theta u + F_\theta w) - z_\theta(g_*), w)$$

is block diagonal with invertible upper block, least singular value at least σ , and lower block the identity. Its derivative norm and its second derivative have common finite bounds on a fixed neighborhood. This follows from Lemma 5.3, finite product rules, and the prefix evidence lower bound κ^n .

Choose a ball inside the command cube on which the derivative differs from its central value by at most half its least singular value. This radius can be chosen uniformly. For a sufficiently small target ball, the map obtained by subtracting the nonlinear remainder and applying the inverse central derivative is a contraction of that command ball into itself. The contraction theorem gives a differentiable inverse. Hence a product of a d -cube and a $(q - d)$ -cube of common positive side length lies in its image, uniformly in θ . All inverse Jacobian determinants are bounded below by a positive constant: the forward derivative is uniformly bounded, so its determinant is uniformly bounded above. The orthogonal command-coordinate change has determinant of absolute value one. No continuity of the chosen orthonormal frames in θ is needed for these uniform bounds.

Uniform commands have density $(1 - 2\eta)^{-q}$. Their joint law with the all-failure word has density

$$(1 - 2\eta)^{-q} \mu \left(\prod_{i=1}^n F_{g_i} \right) \geq (1 - 2\eta)^{-q} \eta^n.$$

Change variables using the inverse just constructed and integrate over the complementary cube. This gives the asserted minorization, with $\beta > 0$ the product of this density bound, the inverse-Jacobian bound, and the two cube volumes. The convention for a zero-dimensional complement is volume one. This is a subprobability bound retaining the failure evidence, not conditioning away a rare prefix. $\square$

5.3. A scale-sensitive finite-state calculation.

Lemma 5.5 (Rectangular quantization at every budget). *For $a_1 \geq \dots \geq a_d > 0$ set*

$$e_M(a) = \max_{1 \leq \ell \leq d} \left(\frac{a_1 \cdots a_\ell}{M} \right)^{1/\ell}.$$

A box with these side lengths has an at-most- M -cell rectangular partition of diameter at most $2\sqrt{d}e_M(a)$ in every cell. The optimal mean squared distortion of its uniform distribution using M arbitrary centers is at least $c_d e_M(a)^2$, with $c_d > 0$ depending only on d . Zero side lengths may be omitted.

Proof. Put $e = e_M(a)$ and take $N_i = \max\{1, \lfloor a_i/e \rfloor\}$ equal intervals in coordinate i . If ℓ coordinates satisfy $a_i \geq e$, then $\prod_i N_i \leq \prod_{i \leq \ell} a_i/e^\ell \leq M$. Every side length in a cell is at most $2e$: for $a_i/e \geq 1$ use $\lfloor a_i/e \rfloor \geq a_i/(2e)$, and otherwise use $a_i < e$. Choose an actual reachable representative in each nonempty cell when covering a reachable subset of the box. The diameter bound remains valid.

For the lower bound project onto the first ℓ coordinates. The projected uniform density is $(a_1 \cdots a_\ell)^{-1}$ on its box. The union of M balls of radius r has probability at most $M v_\ell r^\ell / (a_1 \cdots a_\ell)$, where v_ℓ is the volume of the unit ball. Taking $r = (a_1 \cdots a_\ell / (2M v_\ell))^{1/\ell}$

shows that mean squared distance is at least $r^2/2$. Projection cannot increase distance. Apply this for every ℓ and take the smallest of the finitely many positive dimensional constants. Centers need not lie in the box. $\square$

Proof of Theorem 5.1. Let $D_\theta = \text{diag}(\theta^{\nu_i})$; this temporary matrix notation is unrelated to the peak dimension $D_A(N)$. By Lemma 5.3, $D_\theta z_\theta(h)$ lies in a translated box of side lengths at most $2B\theta^{\nu_i}$ for a fixed B . Lemma 5.5 partitions this box into at most M cells with squared diameter at most $C\Phi_\theta(M)$. Choose a reachable representative in each nonempty cell. At the checkpoint, encode the cell of the exact history and decode its representative's probe probabilities. These are probabilities in $[0, 1]$. Equation (21) gives the uniform upper regret bound. Finite cells with deterministic boundary conventions give Borel encoders.

For the lower bound first average private decoder coins conditional on the message and query. Squared loss cannot increase, leaving at most M centers in probe space. Orthogonally project these centers onto the affine space $p_* + \text{range}(G)$; projection cannot increase distance to a true prediction. Invert G on its range. Its lower singular bound reduces the problem to distance from at most M arbitrary centers in the scaled z coordinates. The law there dominates β times the image of the cube from Lemma 5.4; its side lengths are $2\rho\theta^{\nu_i}$. The lower half of Lemma 5.5 gives $c\Phi_\theta(M)$ after division by the fixed number of queries. Encoder randomization cannot beat distance to the closest center at a fixed history. Fresh public randomness can be fixed and then averaged because it is independent of the entire prefix. At zero, omit zero-weight coordinates in this argument. The same constants, reduced if necessary over the finitely many possible dimensions, work throughout the compact parameter interval. This proves both criteria and the claimed submersion statement. $\square$

Thus the relevant finite-resolution data are the multiplicities of additive collisions and their confluent orders, not only the cardinality of the nearby sumset. The result is a uniform comparison up to constants. It neither asserts a new general quantization principle nor identifies a universal leading distortion constant.

6. FINITE-STATE FILTERING AFTER EVERY REPORT

Fix $N \geq 2$ and abbreviate $d_n = d_A(n, N - n)$ and $D_* = D_A(N) > 0$. The exact state from Theorem 3.4 will be denoted s_n . A clock n , calibration, prior moments and a precomputed finite program are read-only data. The charged resource is the persistent state between successive input symbols.

Definition 6.1 (Finite-state filter). An M -state filter has transitions

$$i_{n+1} = U_n(i_n, g_{n+1}, x_{n+1}), \quad i_n \in \{1, \dots, M\}.$$

It reads only its previous index, the current command and current report. Past commands, past reports and retained command-generation or update seeds are unavailable outside the index. Fresh private coins may be used and then discarded. At a checkpoint the decoder reads only i_n and a newly revealed independent query. The known clock permits stage-dependent transition and output tables. Real command inputs and read-only real calibration constants are allowed; arithmetic workspace, execution time and finite-precision implementation of these fixed functions are not the memory resource of this definition. In particular, this is a finite-state transducer, not an algorithm that repeatedly rereads an exact prefix.

For each $1 \leq n < N$ fix a finite spanning probe list $H_{n,1}, \dots, H_{n,s_n^q}$ as in Proposition 2.3, using $m = N - n$ trials. To avoid confusion, s_n^q is the number of queries, not the exact state s_n . After the checkpoint state is formed, choose an independent uniform query J . Announce

$a \in [0, 1]$, execute that probe using its m fresh trials, and let Y be its all-failure indicator. All N trials are counted. The conditional squared-loss excess over the full-history predictor is

$$(23) \quad \mathcal{R}(a, p) = \frac{1}{s_n^q} \sum_j (a_j - p_j)^2, \quad p_j = \mu(P_h H_{n,j}).$$

This follows by expanding $\mathbb{E}[(a - Y)^2 \mid h, J]$; decoder randomization is averaged on the right. Only one selected checkpoint and one probe are executed in a run. Simultaneous bounds over checkpoints do not count the unexecuted probes as physical trials.

Lemma 6.2 (Uniform finite-horizon transition bounds). *On the compact exact state sets there are constants $L_n < \infty$ such that the update $T_n(s, g, x)$ satisfies*

$$\|T_n(s, g, x) - T_n(\tilde{s}, g, x)\| \leq L_n \|s - \tilde{s}\|$$

for every admitted g, x and $s, \tilde{s} \in \mathcal{S}_n$. Every fixed finite probe prediction vector is likewise Lipschitz in the state, with a finite constant independent of the history.

Proof. In factor coordinates an update before changeover appends the new factor $f/\mu f$. The old factor coordinates are unchanged. All unnormalized report probabilities lie in $[\kappa, 1]$; consequently the normalized factors lie pointwise in $[\kappa, 1/\kappa]$. Their coordinates are bounded, since they are continuous images of a compact cube and finite alphabet. At changeover each needed moment is a ratio of polynomial expressions in these coordinates and the fixed prior moments, with denominator the integral of a product of normalized factors, bounded below by κ^{n+1} . The same lower bound holds on line segments between factor tuples, because each individual interpolated normalized factor is still at least κ . Finite differentiation and the mean value theorem give a uniform derivative bound there.

In moment coordinates the update is (13). Its denominator is at least κ at every state; on a line segment between two such moment vectors the corresponding posterior mixture has the same lower bound. Numerators, coefficients and moment coordinates range over compact sets. The quotient rule therefore gives a uniform bound. Probe predictions are linear in the relevant moment coordinates, or normalized polynomial expressions in the factor coordinates with the same positive denominator control. The argument gives their claimed Lipschitz constants. All constants use only finite algebra and the supplied moments; they may grow with N or deteriorate with κ . $\square$

Theorem 6.3 (Finite-state streaming upper bound). *For every $M \geq 1$ there is a deterministic M -state filter, updated after every report, whose prediction regret at every checkpoint and every admitted command-report history is at most*

$$(24) \quad C_N M^{-2/D_*},$$

where $C_N < \infty$ depends on the fixed experiment, prior, probe lists and total horizon, but not on M or the input history. The representatives and read-only transition functions are supplied existentially from the exact model; this is not an effective synthesis theorem from finite-precision data.

Proof. Enclose $\mathcal{S}_n \subset \mathbb{R}^{d_n}$ in $[-B_n, B_n]^{d_n}$. For $d_n > 0$ partition each coordinate interval into $k_n = \lfloor M^{1/d_n} \rfloor$ intervals. In each nonempty cell choose one representative in $\mathcal{S}_n$. There are at most $k_n^{d_n} \leq M$ representatives, and assigning a state to its cell representative has error at most

$$\delta_n \leq 2B_n \sqrt{d_n}/k_n \leq 4B_n \sqrt{d_n} M^{-1/d_n} \leq CM^{-1/D_*}.$$

Here C is the maximum of the finite prefactors. Empty cells need no representative. For $d_n = 0$ use the unique state. Deterministic boundary and tie conventions give Borel quantizers Q_n .

The filter stores only the index of a representative $\hat{s}_n$. Its transition is exactly

$$(25) \quad \hat{s}_{n+1} = Q_{n+1}(T_n(\hat{s}_n, g_{n+1}, x_{n+1})).$$

Since each representative is reachable and every report has positive probability, its exact image lies in $\mathcal{S}_{n+1}$. Thus the quantizer is always applied inside its domain and no infeasible moment update is used. The finite representative choices and the functions in (25) are read-only program data. They may be obtained by selecting points in nonempty cells of the compact state sets; no efficient global synthesis or computability of unsupplied real prior moments is asserted.

Let $e_n = \|s_n - \hat{s}_n\|$. On the actual command-report stream, Lemma 6.2 gives

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C M^{-1/D_*}.$$

Set $b_0 = 0$ and $b_{n+1} = L_n b_n + C$. Induction proves $e_n \leq b_n M^{-1/D_*}$. This explicitly accumulates the errors from *all* previous lossy updates. At a query use the probe predictions of $\hat{s}_n$, which are actual probabilities. If the vector prediction map has Lipschitz constant K_n , (23) bounds regret by $(K_n^2/s_n^q) b_n^2 M^{-2/D_*}$. The maximum of these finitely many constants proves (24). No exact prefix was kept for the changeover: the factor-to-moment computation uses the representative factors already encoded by i_n . $\square$

Theorem 6.4 (Matching exponent in one fixed streaming experiment). *Choose n_* attaining $D_* = d_A(n_*, N - n_*)$. For the first n_* trials supply independent uniform commands from $\mathcal{G}$, independent of the parameter, cartridges and filtering coins. The realized command is read once by the filter. At n_* reveal the independent probe query and execute its remaining trials. This is one fixed experiment, known before the filter is chosen. Its optimal unconditional regret over M -state filters satisfies*

$$(26) \quad 0 < c_N M^{-2/D_*} \leq \overline{\mathcal{R}}_{M,N}^{\text{stream}} \leq C_N M^{-2/D_*}.$$

The same power is sharp for the minimax streaming criterion taking the maximum conditional regret over checkpoints and input histories. Fresh private update and decoder randomization does not defeat the lower bound; a fresh public coding seed independent of the complete prefix may also be allowed. Thus for fixed N , the required persistent bits for this prediction task satisfy $B_(\varepsilon) = \frac{D_*}{2} \log_2(1/\varepsilon) + O_N(1)$.*

Proof. Write $n = n_*$, $d = D_*$, and $s = s_n^q$. Theorem 3.3 gives an interior command tuple g_0 where the probe map $p(g)$ on all-failure prefixes has rank d . Select d independent probe coordinates, and let E denote their coordinate projection, of operator norm one. Choose complementary command coordinates w so that $g \mapsto (Ep(g), w(g))$ has an invertible derivative at g_0 . Let Φ be its local inverse. There is a compact product $B_\rho^d \times W$ inside its domain, centered at the projected prediction point, such that Φ stays in the command interior and $|\det D\Phi| \geq j_0 > 0$. The complementary box W has positive volume, with unit-volume convention in zero dimensions.

If $k = Kn$ is the number of real command entries, their density is $(1 - 2\eta)^{-k}$. The joint density of commands and an all-failure prefix is this density times $Z(g) = \mu \prod_i F_{g_i} \geq \eta^n$. Change variables with Φ and integrate over W . The subprobability law of the projected predictions on this prefix dominates β times uniform probability on B_ρ^d , with

$$\beta = \frac{\eta^n j_0 |W| |B_\rho^d|}{(1 - 2\eta)^k} > 0.$$

In particular the lower bound retains the probability of the failure prefix. It does not assign an atom to an exact real command or assume a flat ball in the entire curved prediction image.

Every streaming filter is in particular an M -message encoder at this checkpoint. Given any message, average decoder coins conditional on each query; squared loss can only decrease. This

gives at most M prediction vectors. Randomizing the encoder cannot beat the closest of these vectors at a fixed history. Projecting their centers by E cannot increase squared distance. For a uniform point X in a d -ball of radius ρ and any M centers, the union-of-balls estimate gives

$$\Pr\{\min_i \|X - z_i\| \leq r\} \leq \min\{1, M(r/\rho)^d\}.$$

Integrating its complementary tail yields

$$\mathbb{E} \min_i \|X - z_i\|^2 \geq \int_0^{\rho^2 M^{-2/d}} (1 - Mt^{d/2}/\rho^d) dt = \frac{d}{d+2} \rho^2 M^{-2/d}.$$

Multiply by β/s to obtain $c_N > 0$. Fresh public randomness can be fixed and then averaged because it is independent of the prefix. Private online coins may equivalently be drawn on an independent tape for this proof; fixing them makes a deterministic transducer and the same bound holds uniformly. That proof device supplies no tape to the implemented filter. The prescribed exploration is independent of that tape, a condition that is essential to this particular argument. An old command-generation seed is not fresh coding randomness and remains charged.

Theorem 6.3 gives the upper bound in the same experiment. The minimax conditional criterion is at least its unconditional regret, while the universal upper filter works at all checkpoints. Finally put $M = 2^B$ and solve the two inequalities. The Euclidean covering mechanism is classical [4]; the identified sparse dimension and the actual repeated-update construction are the experiment-specific inputs. $\square$

Theorem 6.5 (A separate finite-state control bound). *For the same finite detector and horizon, suppose stage rewards $r_n(g, x) \in [0, 1]$ are continuous in g for each report, and the objective is their expected sum. There is an M -state feedback controller whose loss relative to the full-history optimum is at most*

$$(27) \quad C'_N M^{-1/D*}.$$

No matching lower power for arbitrary control tasks is asserted. An additional finite payoff automaton must be counted as task state, multiplying the number of states if it is included.

Proof. Exact dynamic programming on $\mathcal{S}_n$ is valid by Proposition 2.3. With terminal value zero, its recursion is

$$V_n(s) = \max_{g \in \mathcal{G}} \sum_{x \in \mathcal{X}} p_x(s, g) \{r_{n+1}(g, x) + V_{n+1}(T_n(s, g, x))\}.$$

Here p_x is a linear moment functional or a normalized factor expression. The positive denominator bounds, compact action set and continuous rewards make the displayed objective continuous, and the maximum is attained. By backward induction it is uniformly Lipschitz in s , since finite sums, products of bounded Lipschitz functions, and the updates in Lemma 6.2 preserve that property. The maximum of functions with a common Lipschitz constant is again Lipschitz. Denote by H_n a uniform Lipschitz constant of the action objective, before its maximization.

At each representative choose an exact maximizing action and store that action in the finite output table. At the true state s_n , the loss from using the action optimal at $\hat{s}_n$ is at most $2H_n e_n$. Update the representative by (25). The true exact state is computed only for analysis along the controller's actual observations; the implemented controller still stores only its index. Bellman telescoping and conditional expectation give total loss at most $2 \sum_{n < N} H_n e_n \leq 2 \sum_{n < N} H_n b_n M^{-1/D*}$. This proves (27). It uses no continuity of the optimizing action as a function of state, and it does not differentiate an optimizer. Its approximation principle is related to [8]; it is not claimed as a new general dynamic-programming principle. $\square$

6.1. Horizon and calibration remain part of the constants. If every factor in a probe menu satisfies $0 < F \leq v < 1$, then $H_{n,j} \leq v^{N-n}$ and every posterior prediction satisfies the same bound. The one-state decoder that always announces zero therefore has

$$(28) \quad \mathcal{R} \leq v^{2(N-n)}.$$

For the shared physical menu $F = c + \delta k_j$ one may take $v = c + \delta$. Thus a checkpoint implementation may choose, in advance, between the streaming filter and the one-state decoder and obtain

$$\mathcal{R} \leq \min\{C_N M^{-2/D^*}, (c + \delta)^{2(N-n)}\}.$$

At $c = 1/2$, $\delta = 1/8$, $N - n = 20$, the second bound is $(5/8)^{40} < 10^{-8}$. This does not contradict a fixed- N high-resolution exponent. Neither the new sparse rank theorem nor the streaming construction turns it into a horizon-uniform useful-memory law. We have not rescaled the loss or discarded rare histories to avoid this bound.

The upper constants involve state bounds, transition Lipschitz constants and repeated error propagation. The lower constants involve an attainable Jacobian, a chart radius, evidence and the prior pairing. They are not uniform over vanishing detector amplitudes, all full-support priors or increasing horizons. With a finite fixed command alphabet and finite N there are finitely many histories and an exact finite code eventually suffices; the matching asymptotic lower bound above expressly uses the fixed continuous-command exploration protocol.

7. A UNIFORM STREAMING LAW THROUGH AN ADDITIVE COLLISION

We apply the confluent law to a family in which exact dimension changes at a fixed horizon and at a fixed positivity margin. In this family all repeated quantization errors can be controlled in the physical prediction metric, uniformly at the collision.

7.1. One fixed five-trial protocol. Let $0 \leq \theta \leq 1/2$, $A_\theta = \{0, 1, 2 + \theta\}$, and let μ be any fixed full-support probability on $[0, 1]$. Define

$$(29) \quad \begin{aligned} k_1^\theta &= \frac{1}{3} + \frac{1}{12}t, & k_2^\theta &= \frac{1}{3} + \frac{1}{12}t^{2+\theta}, \\ k_0^\theta &= \frac{1}{3} - \frac{1}{12}(t + t^{2+\theta}). \end{aligned}$$

These cells are at least $1/6$, sum to one, and span W_{A_θ} . Use the fixed comparator cube $[1/4, 3/4]^3$; every report factor is at least $\kappa = 1/24$. The three fixed probe commands have failure factors

$$(30) \quad F_j^\theta = \frac{1}{2} + \frac{1}{8}k_j^\theta, \quad j = 0, 1, 2.$$

The command accepts cell j with probability $3/8$ and the other cells with probability $1/2$. Their sum is $13/8$, so these factors span W_{A_θ} . At a checkpoint n use the 3^{5-n} ordered product queries of length $5 - n$. Exactly one checkpoint and one query are used in a run; all five trials are counted.

For the unconditional experiment the first three commands are independent uniform vectors in the comparator cube. After their reports have been compressed, reveal the independent uniform two-step query. The controller knows θ and μ as read-only calibration, but has access to past commands and reports only through its persistent index. The input law and all query labels are independent of θ .

For a history write $M_s = \mu(P_h t^s)$ and use the five coordinates

$$(31) \quad w = (M_1, M_2, M_{3+\theta}, M_{4+2\theta}, M_{2+\theta} - M_2).$$

They determine every length-two future law. They are ordinary posterior expectations of physical tests and their differences. For analysis put

$$(32) \quad z = (w_1, w_2, w_3, w_4, w_5/\theta) \quad (\theta > 0), \quad z_5|_{\theta=0} = \mu(P_h t^2 \log t).$$

This fifth coordinate at zero is an analytic continuation, not an extra observable supplied to the filter. The zero-calibration future law uses only the first four coordinates.

Theorem 7.1 (Sharp two-parameter streaming order). *Set*

$$(33) \quad \Psi_\theta(M) = \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}, \quad M \geq 1,$$

where the second term is zero at $\theta = 0$. There are constants $0 < c \leq C < \infty$, depending on the fixed prior and protocol but not on M or $\theta \in [0, 1/2]$, with the following properties.

There is a deterministic M -state filter, updated after every report, whose conditional prediction regret at every checkpoint and every admitted history is at most $C\Psi_\theta(M)$. In the prescribed three-trial exploration experiment, every M -state filter has unconditional regret at least $c\Psi_\theta(M)$. The same lower bound therefore holds for the minimax criterion over histories and checkpoints. The lower assertion permits the independent randomization of Definition 6.1.

In particular the two regimes are

$$(34) \quad \overline{\mathcal{R}}_{M,5}^{\theta, \text{stream}} \asymp \begin{cases} M^{-1/2}, & 1 \leq M \leq \theta^{-4}, \\ \theta^{2/5} M^{-2/5}, & M \geq \theta^{-4}, \end{cases}$$

with uniform comparison constants. This is an order-sharp crossover, not a claim of an exact transition budget or a leading distortion constant. The family and its base-model perturbation estimate were introduced in the source-pinned technical note [14]. The conclusion here is the matching, uniform two-parameter law and its online implementation.

7.2. Attainment of the fifth desingularized direction.

Lemma 7.2 (A nondegenerate limiting patch). *At a common interior three-command tuple the map to z in (32) has rank five for every $\theta \in [0, 1/2]$. Its all-failure subprobability law dominates a fixed positive multiple of uniform probability on a translated five-dimensional cube of fixed side length, uniformly in θ .*

Proof. The formal two-fold sum pairs give exponents

$$0, 1, 2, 2 + \theta, 3 + \theta, 4 + 2\theta.$$

There is just one double cluster, at exponent two. Its divided difference is $(t^{2+\theta} - t^2)/\theta$. The nonconstant orders are $(0, 0, 0, 0, 1)$. The condition in (19) is $5 \leq 3(3 - 1) = 6$.

For an explicit common attainable tuple take $c_1 = 1/64$, $c_2 = 2/64$, $c_3 = 3/64$ and failure factors $(1 + c_i t^{2+\theta})/2$. They are realized by rejection probabilities

$$(1 - g_0, 1 - g_1, 1 - g_2) = (1/2 - 2c_i, 1/2 - 2c_i, 1/2 + 4c_i),$$

strictly inside $[1/4, 3/4]^3$. At zero their product tangent is the entire polynomial space of degree at most six. The future confluent space has basis

$$1, \quad t, \quad t^2, \quad t^2 \log t, \quad t^3, \quad t^4.$$

Lemma 5.2 makes its mixed pairing with that tangent surjective. Normalization removes one dimension, not two. For example, for the uniform prior, the minor with rows t^i , $0 \leq i \leq 5$, is exactly

$$(35) \quad \det \left[\frac{1}{i+1}, \frac{1}{i+2}, \frac{1}{i+3}, -\frac{1}{(i+3)^2}, \frac{1}{i+4}, \frac{1}{i+5} \right]_{i=0}^5 = \frac{1}{338751673344000000} > 0.$$

This rational identity is only a diagnostic of the preceding all-prior argument.

For every positive $\theta \leq 1/2$ the six displayed exponents are distinct and the binomial tangent has seven distinct exponents. Strict mixed-moment positivity gives surjectivity throughout this interval. All derivatives in the desingularized coordinates extend continuously to zero, as proved in Lemma 5.3. Compactness and the uniform inverse construction in Lemma 5.4 therefore apply on the *whole* interval $[0, 1/2]$, not only on an unspecified punctured neighborhood. The exact change of order in (32) is a fixed permutation of the general jet coordinates and changes no rank or minorization conclusion. $\square$

7.3. Uniform causal quantization.

Proof of Theorem 7.1. At times one and two retain, exactly for the moment, the chronological normalized factors $f_i/\mu f_i$, with two coefficient coordinates per factor. At time three change to w in (31). At time four retain $(M_1, M_{2+\theta})$, and at time five use the unique terminal state. These states are sufficient for the remaining observable future law. All their reachable sets are compact.

The first two factor sets lie in fixed boxes of dimensions two and four, uniformly in θ . Indeed report coefficients are uniformly bounded, $\mu f \geq \kappa$, and the coefficient hyperplane $\mu f = 1$ can be coordinatized by the two nonconstant coefficients, its constant coefficient then being $1 - \sum_{a>0} f_a \mu(t^a)$. The required prior moments are fixed read-only data for each known calibration. Uniform boundedness of these affine formulas on $[0, 1/2]$ is immediate.

At time three the first four coordinates of w belong to $[0, 1]$. For $0 < t \leq 1$, the fundamental theorem of calculus in the exponent gives

$$|t^{2+\theta} - t^2| \leq \theta t^2 |\log t| \leq \theta.$$

This extends to zero. Consequently $-\theta \leq w_5 \leq 0$ for every posterior, with no evidence denominator in this bound. At time four both retained moments lie in $[0, 1]$. Lemma 5.5, with reachable representatives in nonempty cells, now gives at most M representatives at each time and squared covering errors

$$(36) \quad \delta_1^2 \leq C_1 M^{-1}, \quad \delta_2^2 \leq C_2 M^{-1/2}, \quad \delta_3^2 \leq C_3 \Psi_\theta(M), \quad \delta_4^2 \leq C_4 M^{-1}.$$

The C_i are independent of θ . At zero simply omit the last, constant coordinate of the third state. Every bound in (36) is at most $C_0 \Psi_\theta(M)$.

It remains to verify that repeated updates do not divide by θ . The factor-to-factor update appends the current normalized factor. The factor-to- w update uses the product of the two retained factors and the new report factor. It is a ratio of polynomial coefficient expressions and bounded prior integrals. Along line segments between factor tuples every interpolated normalized factor is at least κ , so the evidence denominators and derivatives have positive uniform bounds. Each physical test in w is uniformly bounded; differentiating in factor coordinates therefore gives a uniform Lipschitz constant for this changeover.

For the only nontrivial shrinking update, write the new report factor as $f = a + bt + ct^{2+\theta}$. In terms of w set

$$Z_f = a + bw_1 + c(w_2 + w_5).$$

The next two moments are exactly

$$(37) \quad M'_1 = \frac{aw_1 + bw_2 + cw_3}{Z_f}, \quad M'_{2+\theta} = \frac{a(w_2 + w_5) + bw_3 + cw_4}{Z_f}.$$

Coefficients are uniformly bounded and $Z_f \geq \kappa$. A line segment between reachable w vectors represents a posterior mixture, so the same lower bound holds along that segment. The quotient

rule proves a uniform Lipschitz estimate in the *physical* w coordinates. There is no divided difference and no factor $1/\theta$ in (37). The next, terminal update is constant.

At each time the filter stores only a representative index and applies the exact transition to that representative and the current input, then requantizes. A reachable representative followed by any admitted report has a reachable image, so every quantizer is used inside its domain. If e_n is the state error in the appropriate Euclidean physical coordinates, the preceding bounds give

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C_0^{1/2} \Psi_\theta(M)^{1/2},$$

with L_n independent of θ . Iterate this recurrence through the five times. At times one and two each query prediction is a normalized polynomial expression in retained factors; at times three and four it is a fixed linear expression in the retained physical moments. These prediction maps have uniform Lipschitz constants, again by the positive evidence bounds. Squaring the resulting prediction error proves the conditional upper bound at every checkpoint. The representatives and transition functions are supplied existentially from the exact model, not synthesized from finite-precision exponent data. Only their index is persistent mutable memory.

For the lower bound at time three apply Lemma 7.2 and the physical-metric argument in the proof of Theorem 5.1, now to the nine common queries in (30). Their coefficient change from any other fixed spanning basis is invertible before taking products, so its full product expansion has constant full column rank. Four physical directions have scale one and the last has scale θ . Lemma 5.5 gives the maximum of the four-dimensional lower bound $cM^{-1/2}$ and the five-dimensional lower bound $c\theta^{2/5}M^{-2/5}$. The constants are uniform on $[0, 1/2]$. Every streaming filter is an M -message encoder at this time, and the retained all-failure evidence is at least $(1/4)^3$. The randomization argument of Theorem 5.1 applies unchanged. Finally, minimax regret is at least the unconditional regret in this one prescribed experiment. This completes the proof. $\square$

Corollary 7.3 (Uniform bit law and the lower-constant scale). *For sufficiently small $\varepsilon > 0$, the least persistent bits for unconditional regret at most ε in the five-trial protocol, and likewise for its checkpoint-uniform minimax criterion, satisfy*

$$(38) \quad B_*(\varepsilon, \theta) = \log_2 \max\{\varepsilon^{-2}, \theta\varepsilon^{-5/2}\} + O(1),$$

where the $O(1)$ is uniform in $\theta \in [0, 1/2]$. If

$$c_{\text{opt}}(\theta) = \inf_{M \in \mathbb{N}, M \geq 1} M^{2/5} \overline{\mathcal{R}}_{M,5}^{\theta, \text{stream}},$$

then $c_{\text{opt}}(\theta) \asymp \theta^{2/5}$ for $0 < \theta \leq 1/2$, with constants independent of θ .

Proof. Solve the two necessary and sufficient inequalities from Theorem 7.1 for M , obtaining, up to constant factors, $M \geq \varepsilon^{-2}$ and $M \geq \theta\varepsilon^{-5/2}$. Rounding to $M = 2^B$ changes B by a bounded amount. The same bounds apply to the minimax criterion. The lower bound for c_{opt} follows from the five-dimensional term. For its upper bound let M tend to infinity in the matching upper estimate, so $M^{-1/10}$ disappears. No limiting leading quantization constant is needed. $\square$

The exact profiles remain $(0, 2, 4, 4, 2, 0)$ at zero and $(0, 2, 4, 5, 2, 0)$ at positive θ . The new assertion is that the extra direction is resolved at distortion scale θ^2 and state-budget scale θ^{-4} , with matching estimates on both sides. It sharpens the base-model bound $C(M^{-1/2} + \theta^2)$ without contradicting it.

8. THE VALUE OF A RESOLVED DIRECTION IN ONE SEQUENTIAL DECISION

We next use the same exploration histories, the same two-step query menu and the same eventual Bernoulli event as in Section 7. Only the terminal decision is changed. This gives a common-payoff comparison between keeping and erasing the newly resolved direction, rather than comparing two separately normalized prediction tasks.

8.1. An independently priced observable ticket. After the time-three state has been formed, reveal the query J and a price U , independent and uniform on $[0, 1]$. The controller chooses a binary action $A \in \{0, 1\}$. Execute the two query trials and let Y be the same all-failure indicator as before. The payoff is

$$(39) \quad A(Y - U).$$

The price and the query are new inputs, not historical storage. The payoff belongs to $[-1, 1]$ and depends only on observable apparatus outcomes and the chosen action. Accepting the ticket changes the payoff, not the detector law or the five-trial budget. The exploration command program is prescribed; it is not optimized by the controller.

Let $p_j(h) = \mathbb{P}(Y = 1 \mid h, J = j)$. Write V_*^θ for the full-history optimal expected payoff and V_M^θ for the supremum over all M -state streaming policies, including randomized transitions and terminal actions. At the terminal decision the policy reads only its index, the new query and the new price.

Proposition 8.1 (Exact decision–prediction identity). *For every fixed streaming encoder with index I , its optimal terminal rule accepts exactly when $U \leq \mathbb{E}[p_J(h) \mid I, J]$. Its value gap is*

$$(40) \quad V_*^\theta - V^\theta(I) = \frac{1}{2s} \sum_{j=1}^s \mathbb{E} \left[(p_j(h) - \mathbb{E}[p_j(h) \mid I])^2 \right], \quad s = 9.$$

Consequently

$$(41) \quad c\Psi_\theta(M) \leq V_*^\theta - V_M^\theta \leq C\Psi_\theta(M)$$

with positive constants independent of M and $\theta \in [0, 1/2]$.

Proof. Conditional on the full history and $J = j, U = u$, action one has expected payoff $p_j - u$ and action zero has payoff zero. The optimum is the threshold rule and its average over price is $\int_0^{p_j} (p_j - u) du = p_j^2/2$. Given only an index, replace p_j by $\bar{p}_{I,j} = \mathbb{E}[p_j \mid I]$, since price and query are independent of the prefix and encoder. A randomized action is a convex combination of the two payoffs and cannot improve the threshold optimum. Thus

$$V_*^\theta = \frac{1}{2s} \sum_j \mathbb{E} p_j^2, \quad V^\theta(I) = \frac{1}{2s} \sum_j \mathbb{E} \bar{p}_{I,j}^2.$$

Conditional-variance decomposition gives (40). Conditional expectation is also the optimal squared-loss decoder for the same encoder. Taking the infimum over the same streaming encoders therefore makes the optimal value gap exactly one half the optimal prediction regret. Theorem 7.1 proves (41). The upper implementation may also threshold the representative's prediction directly: conditional on a history, integrating the mistaken decisions over the interval between that prediction and p_j gives half their squared difference. $\square$

This proposition is a decision identity, not a new general comparison principle. Its use here is to retain the entire resolved streaming law in a bounded, binary-action task and then compare a specific erased observation in that *same* task.

8.2. Erasing the weak coordinate. Let

$$(42) \quad T_\theta(h) = (M_1, M_2, M_{3+\theta}, M_{4+2\theta})$$

be the first four coordinates of (31). This is a deterministic, calibration-dependent postprocessing of the history. It is not a coarsening of a raw one-step detector symbol, and it is not claimed to be the best possible four-dimensional encoding. Let V_T^θ be the optimal payoff of a terminal policy given the *entire exact* statistic T_θ , with no memory restriction, and then the same J, U . This comparison gives the erased controller more computational and storage power than any M -message compression of T_θ .

Theorem 8.2 (Uniform value of the weak direction). *There are constants $c_e, C_e, K > 0$, depending on the fixed prior and protocol but independent of $0 < \theta \leq 1/2$, such that*

$$(43) \quad c_e \theta^2 \leq V_*^\theta - V_T^\theta \leq C_e \theta^2.$$

Moreover, whenever $M \geq K\theta^{-4}$,

$$(44) \quad V_M^\theta - V_T^\theta \geq \frac{1}{2} c_e \theta^2 > 0.$$

In particular an M -state streaming policy beats every policy whose only historical input is an M -message compression of T_θ , even though V_T^θ allows that comparator uncompressed access to the four coordinates. Both values concern the same actual calibration, exploration law, query, price, target, payoff and five-trial budget. At $\theta = 0$ the statistic T_0 is sufficient and $V_^0 = V_T^0$.*

Proof. Put $Z_\theta = w_5/\theta$ for positive θ , with its continuous value at zero as in (32). The common product query expansion has the form

$$(45) \quad p_j(h) = a_j + \ell_j(T_\theta(h)) + \theta b_j Z_\theta(h),$$

where a_j, b_j and the linear forms ℓ_j are independent of θ . They are obtained by expanding the two factors in (30) into the six formal monomials and replacing $M_{2+\theta}$ by $M_2 + \theta Z_\theta$. The full coefficient matrix has rank six, as proved in Lemma 5.3. Hence $\sum_j b_j^2 > 0$; otherwise its weak column would be zero.

Apply the conditional-variance identity from Proposition 8.1 to the sigma-algebra generated by T_θ . With $b_* = (2s)^{-1} \sum_j b_j^2 > 0$ one obtains the exact formula

$$(46) \quad V_*^\theta - V_T^\theta = b_* \theta^2 \mathbb{E} \text{Var}(Z_\theta | T_\theta).$$

The random variable Z_θ is bounded uniformly, since $|(t^{2+\theta} - t^2)/\theta| \leq t^2 |\log t| \leq 1$ before taking any posterior expectation. Thus the conditional variance is uniformly bounded above.

For the lower bound, Lemma 7.2 gives a fixed positive subprobability multiple β of uniform measure on a translated cube of side 2ρ in the coordinates (T_θ, Z_θ) . For every Borel function v of the first four coordinates,

$$\mathbb{E}|Z_\theta - v(T_\theta)|^2 \geq \beta \int |z - v(t)|^2 d \text{Unif}_{\text{cube}}(t, z) \geq \beta \rho^2 / 3.$$

The last inequality integrates the variance of the independent uniform last coordinate; it is unaffected by the translation. Taking the infimum over v proves $\mathbb{E} \text{Var}(Z_\theta | T_\theta) \geq \beta \rho^2 / 3$. This argument uses domination of an actual distribution, not an assumed uniform posterior on a formal tangent space. Equation (46) proves (43) with uniform positive constants.

By (41),

$$V_M^\theta - V_T^\theta \geq c_e \theta^2 - C \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}.$$

For $M \geq K\theta^{-4}$ the maximum on the right is at most $\theta^2 \max\{K^{-1/2}, K^{-2/5}\}$. Choose $K \geq 1$ large enough that $C \max\{K^{-1/2}, K^{-2/5}\} \leq c_e/2$. This proves (44). A policy observing only a compression of T_θ cannot outperform the one observing all of T_θ : conditional on that statistic

its randomization is independent of the latent history, and averaging produces an admissible randomized T_θ -policy. Finally at zero the weak column in (45) vanishes, proving sufficiency and equality of the full and erased values. $\square$

The absolute advantage vanishes quadratically as the exponents collide. What remains nondegenerate is the advantage after normalization by the actual weak-direction scale θ^2 , once the state budget resolves that scale. The theorem is not a lower bound for arbitrary exploration objectives. It is an optimal-policy comparison in one prescribed sequential acquisition experiment, with no change of Bayes baseline and no substitution of the separate mechanical sign experiment.

9. A COLLISION BIT AT A COMMON FINITE-MEMORY BUDGET

This section compares total risks for one fixed target, not regrets against two different Bayes forecasters. It concerns an uncensored raw detector, so that erasure is literal postprocessing. It does not identify this operation with erasure *before* a commanded comparator.

9.1. The counted mechanical realization. Let the triangular lattice have fundamental area $\mathfrak{a} = \sqrt{3}/2$, let $I_R = [9/20, 47/100]$, and put a reflecting disk of radius R at each lattice point. In each cartridge prepare a uniform position in the ambient torus, a uniform velocity direction, and independent uniform apparatus coordinates. A guarded insertion fails when the position is inside a disk. Otherwise a unit-speed particle is released. Every attempted insertion consumes one cartridge.

Fix $0 < T < 3/50$ and $\epsilon > 0$ with $\epsilon(47/100)^2 < 1$. Distinct lifted disks have separation at least $3/50$; an outgoing reflected ray cannot reenter the same convex disk before another collision. Thus each released particle has at most one collision by time T . At a collision let z be its impact coordinate relative to the disk center. The fixed detector reports $+$ when its prepared uniform coordinate is at most $1/2 + \epsilon|z|^2/\pi$, and otherwise reports $-$. The controller never receives z or the uniform seed. This is an internally exact-coordinate apparatus, not a lower bound on the complexity of physical measurement.

For completeness, incoming one-collision states are parameterized by

$$q = Rn_\alpha - sv^-, \quad v^- = -\cos\phi n_\alpha + \sin\phi t_\alpha, \quad 0 < s < T, \quad |\phi| < \pi/2.$$

The backward segment remains outside its own disk by convexity and cannot reach a distinct disk within T by separation. The forward reflected segment has the same exclusion. Every regular colliding initial state has its unique impact triple. At fixed incoming laboratory angle the position Jacobian is $R \cos\phi$, and the angular coordinate change has determinant of absolute value one. Integration against ambient preparation $dq \, d\vartheta/(2\pi\mathfrak{a})$ gives collision mass

$$\frac{R}{2\pi\mathfrak{a}} \int_0^T \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \cos\phi \, d\phi \, d\alpha \, ds = \frac{2TR}{\mathfrak{a}}.$$

Tangencies are null, not an excluded positive-mass neighborhood. Since $|z| = R$ at impact, the four uncensored raw probabilities are

$$(47) \quad \begin{aligned} k_\partial &= \pi R^2/\mathfrak{a}, & k_0 &= 1 - \pi R^2/\mathfrak{a} - 2TR/\mathfrak{a}, \\ k_\pm &= TR/\mathfrak{a} \pm 2T\epsilon R^3/(\pi\mathfrak{a}). \end{aligned}$$

They are positive on I_R and sum to one. For k_0 , positivity follows already at $R = 47/100$, $T = 3/50$, using $\mathfrak{a} > 0.86$ and $\pi < 22/7$; the other inequalities follow from the stated bounds.

Their coefficient matrix in degrees zero through three is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -2T/\mathfrak{a} & T/\mathfrak{a} & T/\mathfrak{a} \\ \pi/\mathfrak{a} & -\pi/\mathfrak{a} & 0 & 0 \\ 0 & 0 & 2T\epsilon/(\pi\mathfrak{a}) & -2T\epsilon/(\pi\mathfrak{a}) \end{pmatrix},$$

with determinant of absolute value $4T^2\epsilon/\mathfrak{a}^3$. A bounded lookup comparator therefore realizes $A = \{0, 1, 2, 3\}$. Merging the signs gives the three probabilities $k_\partial, k_0, k_+ + k_-$, which span $A = \{0, 1, 2\}$. This supplies the dense special cases of Theorem 1.1 from one counted apparatus. The stronger complete-record response, positive filter, boundary-control and surrogate derivations are preserved in the source companion [9]; the collision calculation above uses their whole-preparation normalization rather than conditioning away insertion failures.

9.2. One target and all state budgets. Consume two cartridges without censoring. Let $X \in \{\partial, 0, +, -\}$ be the first raw report, and let Y be the placement-failure indicator on the second cartridge. After the first report the controller stores one of M states, then announces its forecast $a \in [0, 1]$ and incurs $(a - Y)^2$. The erased experiment observes $e(X) \in \{\partial, 0, H\}$ with $e(+)=e(-)=H$. Both experiments use the same prior, both ambient attempts, and the same second-cartridge variable Y . There is no acceptance charge or saved cartridge. This is a separate uncensored experiment; it is not subject to the lower acceptance margin of the streaming theorem.

Put $h(R) = \pi R^2/\mathfrak{a}$ and define

$$p_x = \mu k_x > 0, \quad z_x = \mu(k_x h), \quad b_x = z_x/p_x, \quad p_B = \sum_{x \in B} p_x, \quad z_B = \sum_{x \in B} z_x.$$

For a nonempty subset B set

$$(48) \quad \delta(B) = \sum_{x \in B} \frac{z_x^2}{p_x} - \frac{z_B^2}{p_B} = \frac{1}{p_B} \sum_{\{x,y\} \subset B} p_x p_y (b_x - b_y)^2.$$

All these quantities are finite combinations of prior moments through order five and the exact physical constants.

Theorem 9.1 (Complete common-risk finite-state comparison). *For either alphabet $\mathcal{X}$ (four raw symbols or three erased tags), its optimal M -state risk, allowing independent randomization, is*

$$(49) \quad R_M(\mathcal{X}) = \mu h - \max_{\substack{\mathcal{Q} \text{ partition of } \mathcal{X} \\ |\mathcal{Q}| \leq M}} \sum_{B \in \mathcal{Q}} \frac{z_B^2}{p_B}.$$

A maximizing partition, its block index and block forecasts z_B/p_B give an attained finite-state implementation.

Let R_M^+ and R_M^0 denote the raw and erased risks and $\Gamma_M = R_M^0 - R_M^+$. Then $\Gamma_M \geq 0$ for every M , $\Gamma_1 = 0$, and, with $\Delta = \delta(\{+, -\})$,

$$(50) \quad \Gamma_3 = \Delta - \min_{x \neq y} \delta(\{x, y\}), \quad \Gamma_M = \Delta \quad (M \geq 4).$$

The case $M = 2$ is exactly (49): it compares the seven two-block raw partitions and the three two-block erased partitions (one-block partitions are dominated). Equivalently, if D_M^+ and D_M^0 are the minimum sums of merge costs in the two alphabets, then

$$(51) \quad \Gamma_M = \Delta + D_M^0 - D_M^+.$$

These are common total Bayes risks, not two separately normalized memory regrets.

Proof. Conditional on a block message, the optimal forecast is z_B/p_B . Expanding squared loss gives block contribution $z_B - z_B^2/p_B$; summing proves the risk of every deterministic partition. For arbitrary random encoders and decoders, first replace a randomized forecast at each message by its conditional mean. Holding these at most M forecasts fixed, each raw symbol can be assigned deterministically to a forecast minimizing its conditional squared loss. This cannot increase risk. Reoptimizing each resulting block forecast also cannot increase risk. Thus a deterministic partition is optimal, and the finite maximum is attained. A fresh public seed independent of X, Y merely averages such risks and does not improve the optimum.

The identity (48) is obtained by expanding the weighted variance. The lossless raw risk is $\mu h - \sum_x z_x^2/p_x$; the lossless erased risk exceeds it by Δ . Adding the optimal merging distortion in each alphabet gives (51). Any erased code is a raw code after the deterministic map e , proving $\Gamma_M \geq 0$ directly. With one state both forecasts equal μh . With at least four states both alphabets are losslessly encoded. With three states the erased alphabet is lossless, and an optimal raw partition can be refined to exactly three blocks without increasing risk. Such a partition merges exactly one pair. This proves (50). The partition counts for $M = 2$ are the numbers of nontrivial subsets modulo complementation, namely $(2^4 - 2)/2 = 7$ and $(2^3 - 2)/2 = 3$. $\square$

Write $\mu_j = \mu(R^j)$, let $\mu_H(dR) = R\mu(dR)/\mu_1$, and put

$$v = \text{Var}_{\mu_H}(R^2), \quad \bar{c} = 1/2 + (\epsilon/\pi)\mu_H(R^2).$$

The exact lossless gain retained from [9] is

$$(52) \quad \Delta = \frac{2T\mu_1}{\mathfrak{a}} \frac{(\epsilon v/\mathfrak{a})^2}{\bar{c}(1-\bar{c})} > 0.$$

Indeed conditional on a first hit the sign probability is $c(R) = 1/2 + \epsilon R^2/\pi$, and $\text{Cov}_{\mu_H}(h, c) = \epsilon v/\mathfrak{a}$. The two conditional-mean shifts are respectively this covariance divided by $\bar{c}$ and its negative divided by $1 - \bar{c}$. Their weighted squared sum is the covariance squared over $\bar{c}(1 - \bar{c})$. Multiplication by the hit probability proves (52), which is also precisely the pair cost $\delta(\{+, -\})$. Full support makes $v > 0$. This is an explicit conditional-variance calculation, not a new general comparison-of-experiments principle [1].

Corollary 9.2 (A genuine finite-resolution threshold). *Fix $T > 0$ and a full-support prior on I_R . There exists $\epsilon_* > 0$ such that, for $0 < \epsilon < \epsilon_*$,*

$$(53) \quad \Gamma_M = 0 \quad (M = 1, 2, 3), \quad \Gamma_M = \Delta > 0 \quad (M \geq 4).$$

Thus a four-state, two-bit controller realizes the strict gain, whereas all smaller state budgets have exactly zero gain in this amplitude range. This does not identify exact algebraic dimension with useful accuracy.

Proof. At $\epsilon = 0$, $b_+ = b_- = b_H$. The three distinct tag means satisfy

$$b_0 < \mu h < b_H < b_\partial.$$

The first strict inequality follows because k_0 is strictly decreasing and h strictly increasing, so $\text{Cov}_\mu(k_0, h) < 0$. For example this sign follows by integrating $(k_0(R) - k_0(R'))(h(R) - h(R'))/2$ against two independent prior draws. The second inequality uses the positive covariance of R and h . The last follows in the hit-tilted prior from the same positive covariance of R and h , since the placement posterior is its additional R -tilt. Full support makes all these covariances strict.

For three states the unique zero-distortion merging pattern at amplitude zero is to keep the three distinct tag means separate and merge the two signs. Every other pair has positive merge cost. For two states, no optimal partition at amplitude zero splits the equal-mean signs

between its two blocks. To prove this, use the optimal block means as centers. Equal-mean points assigned to different centers must be equidistant from them; otherwise reassignment to the closer center strictly improves risk. If the centers are distinct, b_H is their midpoint. Each block must then contain a different extreme tag to place its mean strictly on the corresponding side of b_H . Move one sign into the other block, holding centers fixed; the cost is unchanged. The vacated-sign block still contains its extreme tag, so reoptimizing its center strictly reduces the cost, a contradiction. If the two centers agree, the achieved risk is the one-state risk, strictly worse than separating one extreme tag. Thus an optimal two-state partition never splits the signs.

There are finitely many partitions. At amplitude zero every partition that splits the signs and uses at most two states has a strictly positive risk gap to the optimum, as does every non-sign-merging three-state partition. All their risks depend continuously on ϵ because the cell masses stay positive. Therefore, for a sufficiently small common positive ϵ_* , none of these partitions is optimal for $M = 2, 3$. Every remaining partition keeps the signs together and has exactly the risk of the corresponding erased partition, because both $p_+ + p_-$ and $z_+ + z_-$ are independent of ϵ . This proves equality of risks for $M = 2, 3$; $M = 1$ and $M \geq 4$ follow from Theorem 9.1 and (52). $\square$

The threshold is checkable from finitely many moment expressions: enumerate the partitions in (49), find their positive gaps at zero, and bound their continuous rational derivatives while cell masses stay bounded away from zero. No prior-uniform or horizon-uniform threshold is claimed. In particular (52) can be extremely small for a concentrated prior. The threshold theorem gives an exact resource comparison, not a promise of a practically large gain.

10. SCOPE OF THE SYNTHESIS AND RETAINED DERIVATIONS

The central object is the attainable product–test pairing. For sparse monomial detector spaces its rank is classified by Theorem 3.3; the observation algebra determines the future side and the binomial tangent determines the attainable past side. The exact causal state, streaming prediction bounds and finite-state control construction then use that same object. At an additive collision, the confluent pairing supplies additional scale information: its normalized coordinates remain attainable while their physical coefficients vanish at explicit orders. The binary ticket experiment of Section 8 joins this resolved geometry to a common sequential payoff. The mechanical common-risk collision-bit comparison is retained as a separate application, with its own uncensored protocol explicitly specified.

The predecessors [9, 13] are supplied in full in the repository, including every inherited section, proof, foundation file, response and source manifest. Its mechanical experiment retains the whole ambient preparation, placement failure and censoring mass. Its positive likelihood filter, boundary-complete Bellman construction, all-stage acceptance arcs, fixed-controller response, normalized surrogates and acceptance-cost witness have not been removed to obtain the present theorem. They are not counted as newly proved results of the sparse-rank argument.

In particular the fixed-controller response is not differentiation through an optimizing policy, and the surrogate value bound does not preserve exact sparse rank under arbitrary nonpolynomial perturbations. The older acceptance-cost witness is not substituted for the common-risk finite-state comparison in Section 9. The full-posterior control problem with an arbitrary hidden-parameter terminal loss remains different from the observable future-only quotient.

The earlier finite-cylinder response and covariance derivations [11] provide a useful proof discipline: the reporting space, normalizations, derivative envelopes and limiting operation must be stated before a limit theorem is invoked. No long-time, unweighted-seam or many-collision assertion is imported into the present finite-horizon theorem from their historical labels.

Conversely the new finite-horizon results are positive statements for the specified experiment and do not assert a negative conclusion about those separate research directions.

All theorem statements in this manuscript are supported by the written analytic proofs. The accompanying executable diagnostics test finite instances, exact identities, sparse supports, derivative ranks, update arithmetic and partition comparisons. They are neither a proof-assistant certificate nor a journal-level originality or acceptance judgment.

Uniformity in this paper has a specified range. The confluent checkpoint theorem assumes the full-future inequality (19); the matched streaming and binary-decision laws concern the five-trial family on the compact calibration interval $[0, 1/2]$. Their constants may depend on the fixed full-support prior and the physical protocol. There is no claim of a prior-uniform or horizon-uniform lower constant, no assertion of computable synthesis from approximate moment data, and no identification of the history erasure T_θ with a one-step mechanical sign erasure. These distinctions fix the mathematical objects being compared; they do not remove any of the stated positive results.

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